MATH 350-2 Advanced Calculus

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August 31, 2026

Outline

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Completeness Axiom

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An **upper bound** of a set A is a number b with $x \leq b$ for all $x \in A$. If A has an upper bound, it is called **bounded above.**

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Completeness Axiom: every nonempty set of real numbers which is bounded above has a supremum.

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The k'**th decimal apprixomation** of $n.d_0 d_1 d_2 d_3 \ldots$ is

$$s_k = n + \frac{d_1}{10} + \frac{d_2}{10^2} + \frac{d_3}{10^3} + \cdots + \frac{d_k}{10^k}.$$

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Definition

The **decimal** $n.d_0 d_1 d_2 d_3 \ldots$ is the real number defined as the supremum of decimal approximations

$$n.d_0 d_1 d_2 d_3 \cdots = \sup\{s_k : k \in \mathbb{N} \ldots\}.$$

Examples of decimals

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$$0.1100010000000000000000010\ldots$$

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- later we will show *any* positive number has at least one decimal expansion

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In other words

an infimum is a **greatest lower bound**

Properties of Suprema

The first property of suprema is that they must be arbitrarily close to elements of the set.

Theorem (Approximation Property)

Let $S \subseteq \mathbb{R}$ be nonempty and bounded above, and let $b = \sup(S)$. Then for all $\epsilon > 0$, there exists $a \in S$ with

$$b - \epsilon < a \leq b.$$

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Therefore thre must exist $a \in S$ with $a > b - \epsilon$. $\square$

Properties of Suprema

Also, suprema play nicely with addition.

Theorem (Additive Property)

Let $A, B \subseteq \mathbb{R}$ be nonempty, bounded above sets.

$$C = \{x + y : x \in A,\ y \in B\}.$$

Then C is bounded above and

$$\sup(C) = \sup(A) + \sup(B).$$

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Thus $c - a$ is an upper bound of B, and it follows $b \leq c - a$.
Therefore $a + b \leq c$. $\square$

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Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.

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This means there exists $n \in \mathbb{N}$ with $b - 1 < n$.
It follows from Axiom 7 that $b < n + 1$.

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Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.
This means there exists $n \in \mathbb{N}$ with $b - 1 < n$.
It follows from Axiom 7 that $b < n + 1$.
However, $n + 1 \in \mathbb{Z}$, so this contradicts b being an upper bound.

Archimedian property

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Proof.

Replace x with y/x in the previous theorem. □

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- even relations and functions are sets!

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NOTATION:

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n.$$

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Families of sets

A **family of sets** is a collection

$$\{A_i : i \in I\}$$

- I any set, called the **index set**
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Challenge

- index set $I = (1, 2)$
- family of sets $\{A_i : i \in I\}$
- $A_i = [0, 1/i]$

Problem

Determine $\bigcup_{i \in I} A_i$.

Challenge

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The set

$$\text{img}(f) = \{f(a) : a \in A\}$$

is called the **range** or **image** of f